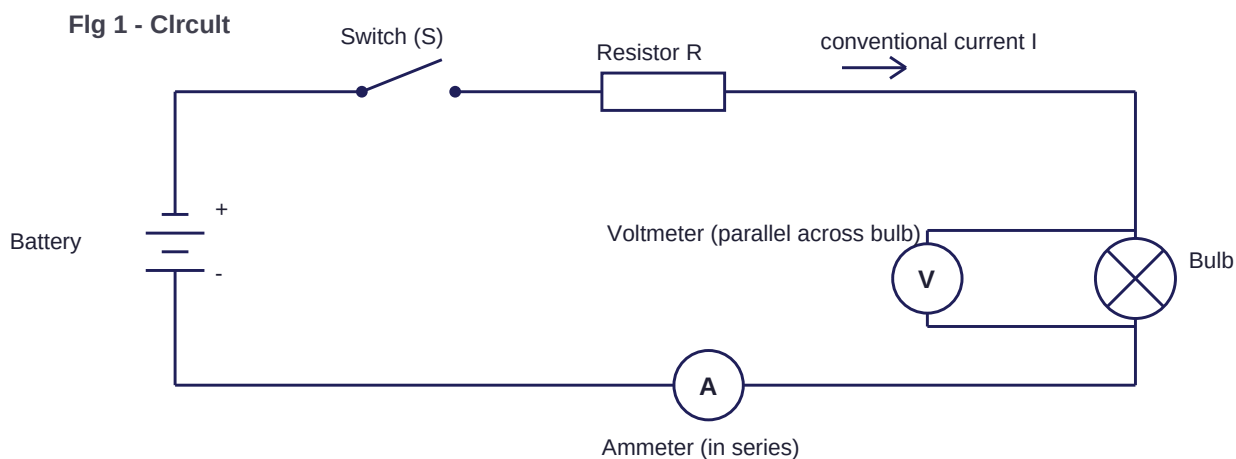


Question 1 — Science (5 marks)

A simple electric circuit is a closed conducting path. The battery gives the potential difference that drives the current round the loop, and the switch opens or closes that path.

The battery, switch, resistor, bulb and ammeter are joined in one loop, that is in series. The ammeter must be in series as the same current passes through every series component. The voltmeter is joined in parallel across the bulb, since it measures the potential difference between the two ends of the bulb.

Conventional current moves from the positive terminal of the battery round the external circuit and back to the negative terminal. By Ohm's law $V = IR$, so raising the resistance at a fixed voltage must lower the current.



|||| scanner edge artefact ||||

Question 2 — English (5 marks)

Technology has made information far easier to reach, but reaching information is not the same as learning from it. That distinction is the whole statement in my view.

It helps most where repetition helps: a student can replay an explanation until it lands, or find a second one when the first does not work. When I could not follow probability in class, three different videos gave me three different angles and one of them finally made sense.

The opposite case is that instant answers remove the struggle, and struggle is where understanding is built. A student who looks up every difficult question finishes the work without learning from it. I think this is a real risk and not a small one.

So my conclusion is that technology raises the ceiling and lowers the floor at once. It rewards the student who wants to understand and it excuses the student who wants to be finished.

Question 3 — Economics (5 marks)

Quantity is on the horizontal axis and price on the vertical axis. Demand slopes downward, supply slopes upward, and they cross at Rs 30 and 60 units, which is the equilibrium because quantity demanded equals quantity supplied there.

Below that price there is a shortage and the price is bid up; above it there is a surplus and the price is cut. A rise in production cost shifts the supply curve to the left, giving a higher price and a lower quantity at the new equilibrium.

Fig 2 - Demand and supply

